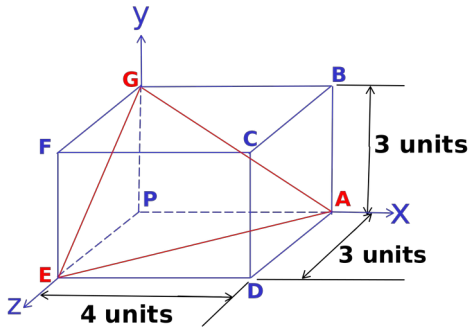
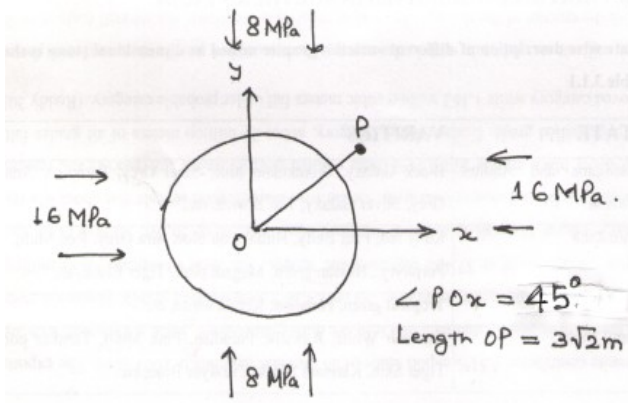


Assignment – 1

Use the conventions of Rock Mechanics for solving the Questions

Q. No.	Question	Marks
1	The state of stress at a point “P” is given by the following tensor $\begin{bmatrix} 5 & 1 & 2 \\ 1 & 6 & 1.5 \\ 2 & 1.5 & 8 \end{bmatrix}$ (a) Determine the magnitude and direction of the principal stresses at the point “P”. (b) Determine the following on the plane $x + 2y + z = 1.6$ - traction vector - magnitude of normal stress - magnitude of the shear stress	5
2	A) Determine the magnitudes and directions of the principal stresses for the following states of stresses. Denote the principal stress state on a suitably rotated stress square. (a) $\tau_{xx} = 25MPa$, $\tau_{yy} = -10MPa$ and $\tau_{xy} = 20MPa$ (b) $\tau_{xx} = -10MPa$, $\tau_{yy} = -20MPa$ and $\tau_{xy} = -20MPa$ B) Verify the results of A on a Graph paper using Mohr’s Circle.	5
3	The strain at a point is given by the matrix $\epsilon = \begin{bmatrix} ax^2 + b & exy & fyz \\ exy & a + by^2 & cxz \\ fyz & czx & dzx \end{bmatrix} \times 10^{-2}$ Find the condition under which the given strain state satisfies the following equations $\frac{2\partial^2 \epsilon_{xy}}{\partial x \partial y} = \frac{\partial^2 \epsilon_{xx}}{\partial y^2} + \frac{\partial^2 \epsilon_{yy}}{\partial x^2}$ $\frac{2\partial^2 \epsilon_{yz}}{\partial y \partial z} = \frac{\partial^2 \epsilon_{yy}}{\partial z^2} + \frac{\partial^2 \epsilon_{zz}}{\partial y^2}$ $\frac{2\partial^2 \epsilon_{zx}}{\partial z \partial x} = \frac{\partial^2 \epsilon_{zz}}{\partial x^2} + \frac{\partial^2 \epsilon_{xx}}{\partial z^2}$ $\frac{\partial^2 \epsilon_{xx}}{\partial y \partial z} = \frac{\partial}{\partial x} \left(-\frac{\partial \epsilon_{yz}}{\partial x} + \frac{\partial \epsilon_{zx}}{\partial y} + \frac{\partial \epsilon_{xy}}{\partial z} \right)$ $\frac{\partial^2 \epsilon_{yy}}{\partial z \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial \epsilon_{yz}}{\partial x} - \frac{\partial \epsilon_{zx}}{\partial y} + \frac{\partial \epsilon_{xy}}{\partial z} \right)$ $\frac{\partial^2 \epsilon_{zz}}{\partial x \partial y} = \frac{\partial}{\partial z} \left(\frac{\partial \epsilon_{yz}}{\partial x} + \frac{\partial \epsilon_{zx}}{\partial y} - \frac{\partial \epsilon_{xy}}{\partial z} \right)$	5

4	The rectangular state of stress at a point 'P' in a rock is given by the array $\begin{bmatrix} 25 & 5 & 3 \\ 5 & 15 & 4 \\ 3 & 4 & 32 \end{bmatrix}$ Determine the magnitude of the normal stress on the plane "AEG" shown in the figure on the right.																	
5	A circular tunnel of 3.0 m radius is proposed to be excavated underground which is subjected to the <i>in situ</i> stress as shown in the figure below.  Determine the principal stresses at point P.	5																
6	a Discuss the Coulomb failure criteria using a neat sketch of Mohr's circle. Then show that in terms of the principal stresses, the criteria can also be expressed as $\sigma_1 = \frac{2c \cos \phi}{1 - \sin \phi} + \sigma_3 \left(\frac{1 + \sin \phi}{1 - \sin \phi} \right)$	7																
	b Show that the slope and intercept of the best fit linear regression line on the σ_1 versus σ_3 graph can be determined using the following set of equations $\begin{pmatrix} m \\ b \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^n \sigma_{3i}^2 & \sum_{i=1}^n \sigma_{3i} \\ \sum_{i=1}^n \sigma_{3i} & n \end{pmatrix}^{-1} \begin{pmatrix} \sum_{i=1}^n \sigma_{3i} \sigma_{1i} \\ \sum_{i=1}^n \sigma_{1i} \end{pmatrix}$	8																
	c A set of 4-triaxial tests were conducted in laboratory. The relevant data are given as under. <table><thead><tr><th>Test No.</th><th>Confining stress (MPa)</th><th>Axial stress at failure (MPa)</th></tr></thead><tbody><tr><td>1</td><td>0</td><td>38.3</td></tr><tr><td>2</td><td>5</td><td>72.4</td></tr><tr><td>3</td><td>15</td><td>115.6</td></tr><tr><td>4</td><td>20</td><td>134.3</td></tr></tbody></table> Determine the effective cohesion and the angle of internal friction	Test No.	Confining stress (MPa)	Axial stress at failure (MPa)	1	0	38.3	2	5	72.4	3	15	115.6	4	20	134.3	5	
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7	Three flat Jack tests have been carried out inside a raise in a metal mine. The centre line of the raise dips 70° towards West. The slots were cut normal to the wall of the raise and oriented relative to the centre line as shown in figure. The cancellation pressures in each of the Flat Jacks are given as under. <table><tr><th>Flat Jack</th><th>Cancellation Pressure (MPa)</th></tr><tr><td>A</td><td>18.00</td></tr><tr><td>B</td><td>15.52</td></tr><tr><td>C</td><td>17.30</td></tr></table> Assume that the cancellation pressure is equal to the normal stress acting across the flat jack slot. The pre-mining state of stress in the x-y co-ordinate system is _____	Flat Jack	Cancellation Pressure (MPa)	A	18.00	B	15.52	C	17.30	5
Flat Jack	Cancellation Pressure (MPa)									
A	18.00									
B	15.52									
C	17.30									
8	Stress measurement using door stopper is carried out. The sensor uses a 60° strain rosette (angle between A & B as well as between B&C both are 60°). The strain gauge A is aligned along the x-direction while fixing at the end of the borehole wall. The maximum strain recorded by the Strain gauges A, B and C are 1200, 500, 500 micro strains respectively. The constants in the matrix relating the far field stress and the stresses at the end of the borehole wall are $a = 1.25$, $b = -0.064$, $c = -0.75(0.645 + \nu)$, $d = 1.25$. Given that the Young's Modulus of the rock is $E = 5 \text{ GPa}$ and the Poisson's ratio is $\nu = 0.30$. The state of pre-mining stress in 2D is _____	5								
9	A circular shaft of 5.0 m diameter is being excavated for mining. At a certain depth the shaft section is subjected to in situ stress as shown in the figure. The failure of the rock mass is governed by the failure criteria $\sigma_1 = \sigma_3 + \sqrt{m\sigma_c\sigma_3 + \sigma_c}$ where, $\sigma_c = 50 \text{ MPa}$ and $m = 7$. The factor of safety of the point 'P' shown in figure is _____ Use the Mohr circle concept to find the FoS.	5								
10	While conducting hydraulic fracturing test in hard and impermeable rock of tensile strength 6.8 MPa, the re-fracturing pressure was recorded as 6.2 MPa and the shut-in pressure was recorded as 5.2 MPa. The magnitude of maximum horizontal stress is _____	5								

